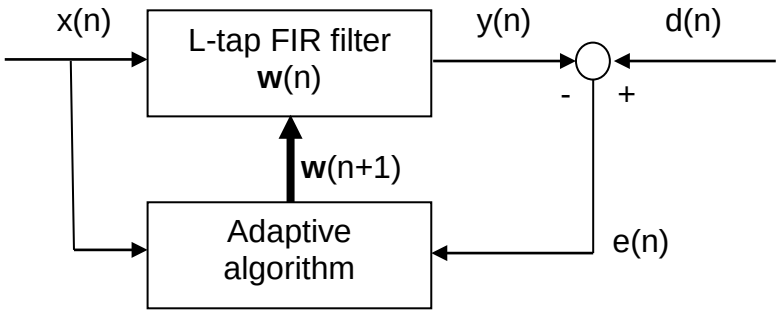
Optimal adaptive filter – Wiener Hopf



Cost-function

$$\xi(n) = \mathrm{E}[e^2(n)] = \mathrm{E}\left[\left(d(n) - \mathbf{w}^{\mathrm{T}}(n)\mathbf{x}(n)\right)^2\right]$$
$$= \mathrm{E}[d^2(n)] - 2\mathbf{w}^{\mathrm{T}}(n)\mathrm{E}[d(n)\mathbf{x}(n)] + \mathbf{w}^{\mathrm{T}}(n)\mathrm{E}[\mathbf{x}(n)\mathbf{x}^{\mathrm{T}}(n)]\mathbf{w}(n)$$

Autocorrelation matrix

$$\mathbf{R}_{xx}(n) = \mathrm{E}[\mathbf{x}(n)\mathbf{x}^{\mathrm{T}}(n)] = \begin{bmatrix} r_{xx}(0) & \cdots & r_{xx}(L-1) \\ \vdots & \ddots & \vdots \\ r_{xx}(L-1) & \cdots & r_{xx}(0) \end{bmatrix}$$

Cross correlation vector

$$\mathbf{r}_{dx}(n) = \mathrm{E}[d(n)\mathbf{x}(n)] = \begin{bmatrix} r_{dx}(0) \\ \vdots \\ r_{dx}(L-1) \end{bmatrix}$$

Giving

$$\xi(n) = \mathrm{E}[d^2(n)] - 2\mathbf{w}^{\mathrm{T}}(n)\mathbf{r}_{dx}(n) + \mathbf{w}^{\mathrm{T}}(n)\mathbf{R}_{xx}(n)\mathbf{w}(n)$$

Optimization

$$\boldsymbol{\nabla}\xi(n) = \frac{\partial \xi}{\partial \mathbf{w}} = 0 - 2\mathbf{r}_{dx}(n) + 2\mathbf{R}_{xx}(n)\mathbf{w}(n) = \mathbf{0}$$

Optimal filter (Wiener Hopf solution)

$$\mathbf{w}(n) = \mathbf{w}_{opt}(n) = \mathbf{R}_{xx}^{-1}(n)\,\mathbf{r}_{dx}(n)$$